Federated Gradient-Refined Adaptive Averaging for Intrusion Detection Systems

No Author Given

No Institute Given

Abstract. Federated learning has become a powerful framework for intrusion detection systems by enabling collaborative learning across distributed clients while maintaining data privacy. However, the standard *FedAvg* approach, encounters significant limitations when handling heterogeneous data distributions among clients. In cases where certain clients have substantially less data, their contributions to the global model are disproportionately reduced, leading to suboptimal performance, especially in detecting classes with fewer samples. To overcome this issue, we propose a novel method called `FedGRAA` which employs an adaptive, gradient-based weight update mechanism that dynamically adjusts the significance of each client in each training round based on the distribution of their local data. This weighting mechanism ensures that clients with smaller datasets still have a meaningful influence on the global model, thereby enhancing detection accuracy for underrepresented classes. Extensive experiments on CICIDS 2017 and UNSW-NB15 demonstrate that `FedGRAA` achieves 88% accuracy on minority classes, significantly outperforming *FedAvg* while preserving client privacy. Additionally, as the number of clients with minority samples increases, `FedGRAA` maintains its advantage, outperforming *FedAvg* by up to 55%. This marks a major advancement in federated learning for intrusion detection, especially in non-IID settings.

1 Introduction

The rapid growth of network technologies and the increasing number of connected devices have escalated cybersecurity risks, making Intrusion Detection Systems (IDS) crucial for detecting and preventing attacks. Traditional IDS typically depend on centralized data collection, which raises concerns about privacy and limited adaptability [1]. Federated Learning (FL) introduces a decentralized solution, enabling multiple parties to collaboratively train models without sharing raw data. This approach safeguards privacy and enables faster model updates, ensuring an agile response to new intrusions.

However, FL faces significant challenges, particularly due to heterogeneous data distributions across clients. Federated Averaging (*FedAvg*), introduced by McMahan et al. [6], allows privacy-preserving model training but assumes homogeneity in client data—a condition that rarely holds in real-world scenarios. Data heterogeneity, also referred to as non-IID (non-independent and identically

distributed) data, is especially problematic in federated IDS applications, where different clients encounter distinct types of attacks at varying frequencies. For instance, certain clients may face frequent attempts of one type of intrusion, while others rarely see it. Such variations can cause conventional FL approaches to yield biased and less effective models [3].

In response to these challenges, adaptive federated learning frameworks have been proposed. Zhang et al. [12] introduced *FedSND*, which dynamically adjusts to data disparities by modifying aggregation weights based on data distribution statistics. Despite these advancements, most adaptive techniques continue to struggle with highly imbalanced datasets in IDS, where rare but critical attack patterns are underrepresented. This imbalance leads to inadequate detection of such patterns, heightening security risks.

To tackle this issue, we propose a novel method, *Federated Gradient-Refined Adaptive Averaging* (`FedGRAA`), which enhances model performance for clients with underrepresented data by dynamically adjusting aggregation weights based on data distribution and gradient updates at each training round. This approach refines the aggregation process, granting higher weight to gradients from clients with rare attack patterns. In our evaluation using the imbalanced CIC-IDS-2017 dataset, `FedGRAA` outperforms both traditional federated averaging and static reparameterization methods, achieving notable improvements in detection accuracy for underrepresented attack types. This advancement in adaptive FL approaches paves the way for more robust and responsive IDS solutions in federated environments.

2 Related Work

Static Reparameterization Static parameter aggregation methods in federated learning struggle with non-IID data. Zhao et al. [13] and Li et al. [5] demonstrated how *FedAvg* suffers performance degradation and poor convergence in heterogeneous environments due to its static nature. Fallah et al. [2] and Reisizadeh et al. [8] further highlighted these limitations, proposing personalized federated learning and meta-learning techniques as solutions. These studies show that static reparameterization is inadequate for handling client diversity, underscoring the need for adaptive methods.

Static reparameterization lacks adaptability to changing data distributions, leading to model rigidity and bias towards initial conditions (Smith et al. [5]). This approach underrepresents crucial but rare data, affecting applications like anomaly detection and medical diagnostics (Pranab et al. [9]). Furthermore, static models fail to capture the dynamic nature of real-world data, limiting long-term relevance (Lee & Kim [4]). This highlights the need for dynamic strategies that adapt to new data without manual intervention (Zhao & Joshi, 2022 [14]).

Adaptive Reparameterization This overcomes the limitations of static reparameterization by dynamically adjusting aggregation weights based on their performance throughout the training process. Unlike static methods, where client contributions remain fixed, adaptive reparameterization recalibrates the influence

of each client to ensure the global model remains balanced, even in non-IID settings. This technique, seen in methods like *FedOpt* [7] and *FedProx* [5], reduces bias towards clients with majority-class data, emphasizing underrepresented clients, and enhancing model generalization.

Our method (`FedGRAA`) introduces an adaptive reparameterization approach tailored for the non-IID nature of intrusion detection tasks. By continuously adjusting aggregation weights based on gradients, the method ensures that both common and rare data are adequately represented. This dynamic adjustment leads to a robust model capable of detecting rare attack types like *Heartbleed*, *SQL Injection*, *Infiltration* (CIC-IDS-2017) in intrusion detection systems. As a result, our approach outperforms *FedAvg* [6] by achieving more balanced and effective performance in heterogeneous data environments, particularly in handling rare and critical attack types that *FedAvg* [6] struggles to detect.

3 Background and Problem Formulation

Federated Learning It is a distributed framework that trains a global model across decentralized devices without sharing local data, preserving privacy and reducing communication costs. The most common algorithm, Federated Averaging (*FedAvg*), works by initializing global model parameters θ_g^0, selecting a fixed set of clients S_t at each round t, and sending the current model θ_t to them: $\theta_k^{t+1} \leftarrow \theta_g^t$. Clients update their local models using stochastic gradient descent (SGD) on their local datasets D_k:

$$\theta_k^{t+1} = \theta_k^{t+1} - \eta \nabla L_{\theta_k^{t+1}}(\theta_k^{t+1}; D_k), \quad \forall k \in S_t \tag{1}$$

where η is the learning rate and K is the total number of clients in S_t. The server then aggregates these updates to form the global model and this process repeats until convergence [6][10]:

$$\theta_g^{t+1} = \frac{1}{K} \sum_{k \in S_t} \theta_k^{t+1} \tag{2}$$

Non-IID Data Challenges in Federated Learning In Federated Learning, data is often non-independent and identically distributed across clients, leading to challenges such as bias, where the global model may overfit to certain clients' data, and slower convergence, requiring more communication rounds [13]. The loss function for client k is defined as:

$$L_k(\theta) = \frac{1}{|D_k|} \sum_{i \in D_k} l(\theta; x_i, y_i) \tag{3}$$

where $l(\theta; x_i, y_i)$ is the loss for a single data point (x_i, y_i), and $|D| = \sum_{k=1}^{K} |D_k|$ is the total number of data points. The global loss function $L(\theta)$ is the weighted average of local loss functions:

$$L(\theta) = \sum_{k=1}^{K} \frac{|D_k|}{|D|} L_k(\theta) \tag{4}$$

In Federated Averaging, a client's contribution to the global model is proportional to its dataset size, meaning clients with fewer samples have less influence, as seen in the aggregation step [6]:

$$\theta_g^{t+1} = \sum_{k \in S_t} \frac{|D_k|}{|D|} \theta_k^{t+1} \tag{5}$$

Non-IID data can also slow convergence as gradients from different clients may diverge. When gradients $\nabla L_k(\theta)$ are misaligned, the global gradient $\nabla L(\theta)$ may oscillate, delaying convergence. Techniques like personalization, regularization, and adaptive aggregation address these issues [3]:

$$\nabla L(\theta) = \sum_{k=1}^{K} \frac{|D_k|}{|D|} \nabla L_k(\theta) \tag{6}$$

With non-IID data, clients with fewer samples have limited influence on the global model due to the weighting factor $\frac{|D_k|}{|D|}$ in the *FedAvg* algorithm. This under-representation prevents smaller clients from receiving models tailored to their local data. Consequently, the aggregation process favors clients with larger datasets, leading to a global model biased towards them and reducing performance on tasks for smaller clients, i.e., the global model θ_g^{t+1} tends to approximate θ_j^t for large $|D_j|$.

4 Methodology

Static Reparameterization In the static reparameterization approach below, we assign fixed weights to each client's contribution based on the inverse of their dataset size. This is mathematically represented as follows:

$$\theta_g^{t+1} = \sum_{k=1}^{K} \frac{1/|D_k|}{\sum_{j=1}^{K} \frac{1}{|D_j|}} \theta_k^t \tag{7}$$

where $|D_k|$ denotes the number of samples for client k. In this formulation, clients with smaller datasets have a proportionally larger influence on the global model, as contributions are weighted inversely to dataset size. Although this method helps to mitigate the dominance of large datasets, it lacks the flexibility to adapt to variations in data distribution or client performance over time.

Adaptive Reparameterization (`FedGRAA`) We introduce an adaptive reparameterization method, `FedGRAA` (Algorithm 2), which dynamically adjusts aggregation weights based on performance changes, with a focus on prioritizing minority classes. The adjustment is driven by a personalization co-efficient γ, where $\gamma \in [0, 1]$. As γ approaches 1, the method prioritizes minority classes, while decreasing γ towards 0 reduces their emphasis, allowing for effective handling of data heterogeneity in federated learning, particularly in non-IID settings.

The process begins with the server initializing the global model parameters, θ_g^0, while the initial aggregation weights, $w^0 = [w_1^0, \ldots, w_K^0]$, are calculated via static reparameterization (Equation 7). During each training round t, a fixed set of clients S_t is selected to participate, and the global model parameters θ_g^{t-1} are sent to each client. Each client k initializes its local model $\theta_k^t \leftarrow \theta_g^{t-1}$ and performs updates using gradient descent on local data batches from D_k for E epochs. Once local training is complete, the updated model parameters, θ_k^t, are sent back to the server (Algorithm 1)

Algorithm 1 Adaptive Reparameterization for Model Personalization

Input: T - Training rounds, η & η' - learning rates, $w^0 = [w_1^0, w_2^0, \ldots, w_K^0]$ - initial aggregation weights computed via static reparameterization (Equation 7), γ - personalization coefficient, $\mathcal{D}_\mathcal{V}$ - validation dataset, $\mathcal{D}_\mathcal{V}^n$ - validation dataset of last n underrepresented samples.

Server executes:
  initialize global model parameters θ_g^0
  for each round $t = 1, 2, \ldots, T$
    $S_t \leftarrow$ (fixed set of K clients)
    for each client $k \in S_t$ **in parallel**
      send current global model θ_g^{t-1} to client k
      $\theta_k^t \leftarrow \textsc{ClientUpdate}(k, \theta_g^{t-1})$
    $\theta_c^t = [\theta_1^t, \theta_2^t, \ldots, \theta_K^t]$
    $w^t \leftarrow \textsc{FedGRAA}(\theta_c^t, w^{t-1})$
    $\theta_g^t = \sum_{k \in S_t} w_k^t \theta_k^t$ // update global model
  return θ_g^T // Final global model parameters

$\textsc{ClientUpdate}(k, \theta_g^{t-1})$: // Run on client k
  initialize local model $\theta_k^t \leftarrow \theta_g^{t-1}$
  for each local epoch from 1 to E
    for each batch b in D_k // each batch of size B
      $\theta_k^t \leftarrow \theta_k^t - \eta \nabla L_{\theta_k^t}(\theta_k^t; b)$
  return θ_k^t to server

Once the updated local models are received from the clients, the server computes an intermediate global model, θ_g', by using the weights from the previous round, w^{t-1}: $\theta_g' = \sum_{i=1}^m w_i^{t-1} \theta_i^t$.

We split the dataset $\mathcal{D}_v \setminus \mathcal{D}_v^n$ into two parts, $\mathcal{D}_\mathcal{V}^h$ and $\mathcal{D}_\mathcal{V}^m$, using an automated method that determines the split threshold based on the median frequency. To refine the weight adjustment process further, the aggregation weights are updated by optimizing a combined loss function L' that incorporates both the loss L_u, which focuses on $\mathcal{D}_v^n$, and L_a, which is a function of losses defined for highly and moderately frequent data. The L_a component is defined as follows:

$$L_a = \min(L_h(\theta_g'; \mathcal{D}_\mathcal{V}^h), L_m(\theta_g'; \mathcal{D}_\mathcal{V}^m)) \tag{8}$$

Here, θ_g' is the temporary global model used to update the aggregation weights.

This combined loss is expressed as:

$$L'(\theta'_g; \mathcal{D}_\mathcal{V}, \mathcal{D}^n_\mathcal{V}) = (1-\gamma)L_a(\theta'_g; \mathcal{D}^h_\mathcal{V}, \mathcal{D}^m_\mathcal{V}) + \gamma L_u(\theta'_g; \mathcal{D}^n_\mathcal{V}) \tag{9}$$

Here, $\mathcal{D}_\mathcal{V}$ represents the whole validation dataset, while $\mathcal{D}^n_\mathcal{V}$ refers to the validation data for the last n underrepresented classes. The parameter γ controls the balance between prioritizing overall performance and improving accuracy for these underrepresented classes (Equation 9).

Since θ'_g is a function of w^{t-1}, we take the gradient of the loss function with respect to w^{t-1}. The weights w^{t-1} are then updated by minimizing the gradient of this combined loss function:

$$w^{t-1} \leftarrow w^{t-1} - \eta' \nabla_{w^{t-1}} L'(\sum_{i=1}^{K} w_i^{t-1}\theta_i^t; \mathcal{D}^h_\mathcal{V}, \mathcal{D}^m_\mathcal{V}, \mathcal{D}^n_\mathcal{V}) \tag{10}$$

After completing this weight update for E' epochs, the new weights $w^t \leftarrow w^{t-1}$ are applied, and at the end of the tth round, the global model is updated as: $\theta^t_g = \sum_{k \in S_t} w^t_k \theta^t_k, \forall k \in S_t$.

Algorithm 2 Federated Gradient-Refined Adaptive Averaging (`FedGRAA`)

FedGRAA(θ^t_c, w^{t-1}): *// Our Algorithm*
  for each local epoch from 1 to E'
    $\theta'_g = w_1^{t-1}\theta_1^t + w_2^{t-1}\theta_2^t + \cdots + w_K^{t-1}\theta_K^t$
    $L_a = \min(L_h(\theta'_g; \mathcal{D}^h_\mathcal{V}), L_m(\theta'_g; \mathcal{D}^m_\mathcal{V}))$
    $L' = (1-\gamma)L_a(\theta'_g; \mathcal{D}^h_\mathcal{V}, \mathcal{D}^m_\mathcal{V}) + \gamma L_u(\theta'_g; \mathcal{D}^n_\mathcal{V})$
    $w^{t-1} \leftarrow w^{t-1} - \eta' \nabla_{w^{t-1}} L'(\theta'_g; \mathcal{D}^h_\mathcal{V}, \mathcal{D}^m_\mathcal{V}, \mathcal{D}^n_\mathcal{V})$
  $w^t \leftarrow w^{t-1}$
  return w^t

`FedGRAA` dynamically adjusts aggregation weights using the personalization coefficient γ, prioritizing minority classes to address data heterogeneity. Unlike static reparameterization with fixed weights, `FedGRAA` adapts to varying client contributions, ensuring better generalization and faster convergence (see Section A.1 for convergence proof) in non-IID settings. Studies [7] [11] show that adaptive reparameterization enables personalized updates, reducing outlier's influence and improving model convergence and performance in federated learning.

5 Experiments and Results

Data Settings In our federated learning setup, the datasets (see Tables 1 and 2) are divided among clients. To prevent highly frequent attacks, such as *DoS Hulk*, *PortScan*, *DDoS*, *DoS GoldenEye* (CIC-IDS data) and *Generic*, *Fuzzers*, *Reconnaissance* and *DoS* (UNSW data), from dominating, their samples are capped at 10,000. Each client is assigned data for one specific attack type paired with an equal number of *Benign* (*Normal* for UNSW data) samples to ensure balanced training. Clients use 90% of their data for local training and reserve 10% for validation ($\mathcal{D}_\mathcal{V}$), enhancing detection across both frequent and rare attacks.

Network Architecture We designed a fully connected neural network using Keras Sequential API for our intrusion detection model. The network accepts 78 and 43 input features for CIC-IDS-17 and UNSW datasets, has hidden layers of sizes 200, 100, and 50 with *tanh* activation, and an output layer of 15 and 10 neurons for CIC-IDS-17 and UNSW datasets with *softmax* activation, corresponding to their respective classes in the CIC-IDS-2017 and UNSW-NB15 datasets.

Training Setup and Hyperparameters The model was trained using the Adam optimizer with a learning rate of $\eta = 0.01$. Each client performed local training for $E = 2$ epochs per round with a batch size of $B = 64$. The federated learning process was conducted over 200 communication rounds with $K = 14$ and $K = 9$ clients participating for CIC-IDS-17 and UNSW datasets respectively. The loss function used was sparse categorical cross-entropy.

ID	Label	Original	Used
0	BENIGN	2,273,097	63,901
1	Bot	1,966	1,948
2	DDoS	128,027	10,000
3	DoS GoldenEye	10,293	10,000
4	DoS Hulk	231,073	10,000
5	DoS Slowhttptest	5,499	3,219
6	DoS Slowloris	5,796	5,228
7	FTP-Patator	7,938	5,931
8	Heartbleed	11	11
9	Infiltration	36	36
10	PortScan	158,930	10,000
11	SSH-Patator	5,897	5,385
12	Brute Force	1,507	1,470
13	SQL Injection	21	21
14	XSS	652	652

Table 1: Original and used sample distribution for CIC-IDS-17 data.

ID	Label	Original	Used
0	Normal	93,000	56,691
1	Generic	58,871	10,000
2	Fuzzers	24,246	10,000
3	Backdoor	2,329	2,329
4	Worms	174	174
5	Reconnaissance	13,987	10,000
6	Analysis	2,677	2,677
7	Exploits	44,525	10,000
8	Shellcode	1,511	1,511
9	DoS	16,353	10,000

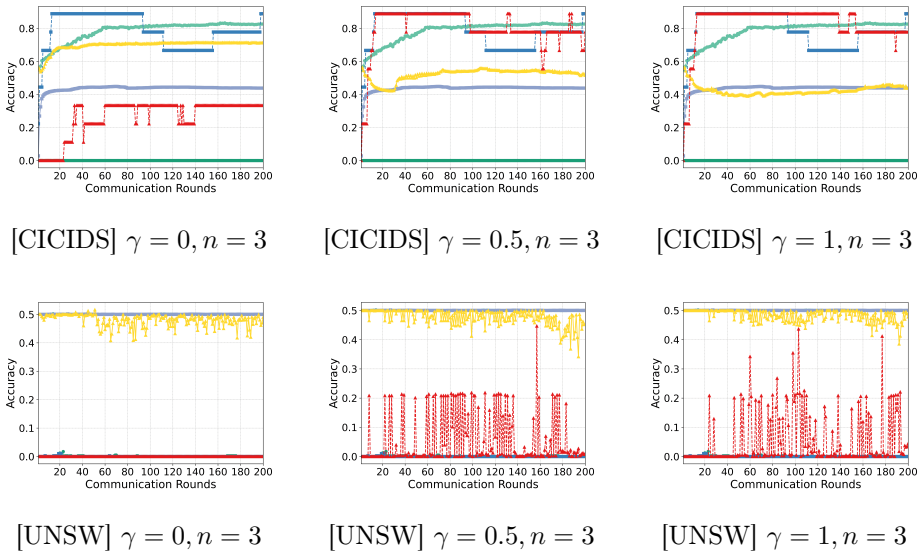
Table 2: Original and used sample distribution for UNSW-NB15 data.

For `FedGRAA`, we continued using the Adam optimizer, with a learning rate $\eta' = 0.1$ and epochs $E' = 5$. Different values of the personalization coefficient γ (0, 0.5, 1) were experimented with to adjust the influence of clients with underrepresented data, focusing on the three ($n = 3$) underrepresented clients with rare attack types: Heartbleed, SQL Injection, and Infiltration for CIC-IDS dataset and Worms, Shellcode and Backdoor for UNSW dataset.

Algorithm Comparison We compared the performance of our proposed method, `FedGRAA`, against the standard *FedAvg* and Static Reparameterization methods (see Section 4). Figure 1 shows the accuracy trends over 200 communication rounds for different γ values (0, 0.5, 1).

Results and Analysis The results in Figure 1 demonstrate that our approach balances accuracy for underrepresented classes and overall global accuracy by tuning γ. As γ increases, accuracy for minority classes (e.g., Classes [8, 9, 13]

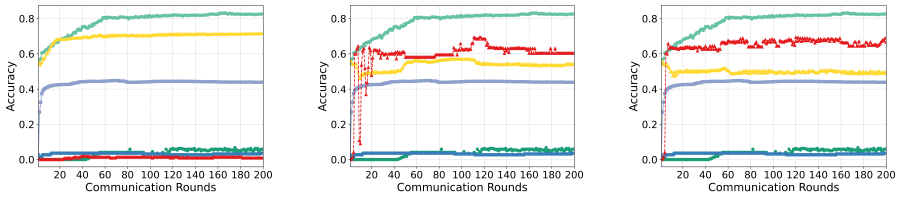
in CICIDS, [4, 8, 3] in UNSW) improves, with a minor reduction in global accuracy. For higher values ($\gamma = 1$ and $\gamma = 0.5$), `FedGRAA` excels in minority class performance while maintaining global accuracy comparable to or better than Static Reparameterization and *FedAvg*. Conversely, at $\gamma = 0$, `FedGRAA` prioritizes global accuracy, outperforming Static Reparameterization, but sacrifices minority class performance by deprioritizing these classes.



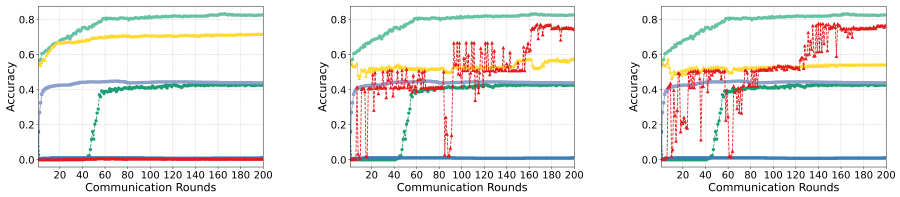
Method	**Acc. (CIC-IDS-17)**	**Acc. (UNSW-NB15)**
FedAvg	00.00%	00.00%
Static Reparam.	66.66%	00.00%
FedGRAA ($\gamma = 1$)	**88.88%**	**21.09%**
FedGRAA ($\gamma = 0.5$)	**77.77%**	**20.59%**
FedGRAA ($\gamma = 0$)	33.33%	00.00%

Fig. 1: Performance comparison for varying γ values on CIC-IDS and UNSW.

Figure 2 evaluates the robustness of our FedGRAA algorithm as the number of clients participating in personalization increases. For $n = 5$ and $\gamma = 1$, FedGRAA achieved an accuracy of approximately 65%, significantly outperforming FedAvg and static reparameterization, which achieved accuracies below 10%. Similarly, for $n = 7$ and $\gamma = 1$, FedGRAA achieved around 80% accuracy, compared to 40% for FedAvg and just 2% for static reparameterization. Comparable results are observed in Figure 3 for the UNSW dataset, with similar trends for $n = 5$ and $n = 7$. These findings confirm FedGRAA's consistent robustness as the number of clients with minority samples increases.

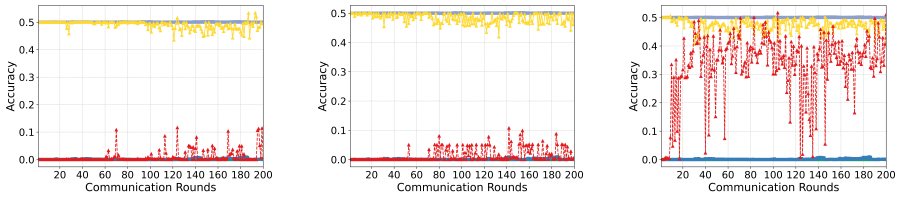


(a) $\gamma = 0, n = 5$ (b) $\gamma = 0.5, n = 5$ (c) $\gamma = 1, n = 5$

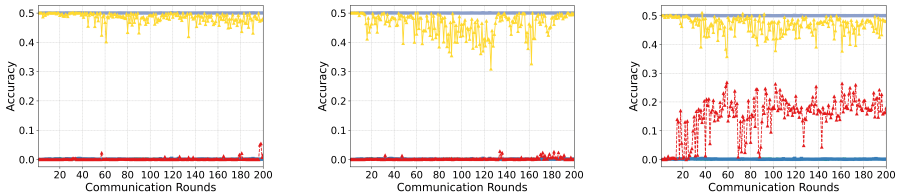


(d) $\gamma = 0, n = 7$ (e) $\gamma = 0.5, n = 7$ (f) $\gamma = 1, n = 7$

Fig. 2: Performance comparison for varying γ and n values for CIC-IDS-17 dataset.



(a) $\gamma = 0, n = 5$ (b) $\gamma = 0.5, n = 5$ (c) $\gamma = 1, n = 5$



(d) $\gamma = 0, n = 7$ (e) $\gamma = 0.5, n = 7$ (f) $\gamma = 1, n = 7$

Fig. 3: Performance comparison for varying γ and n values for UNSW dataset.

6 Conclusion

We proposed a federated learning framework for intrusion detection systems that tackles the challenge of data heterogeneity through `FedGRAA`. Our approach dynamically adjusts aggregation weights, ensuring that clients with rare attack data have a greater impact on the global model, thus reducing bias towards more

common patterns like benign traffic. This adaptive weighting improves detection accuracy for underrepresented attacks, as demonstrated on the CIC-IDS-2017 and UNSW-NB15 datasets. In future work, we aim to develop adaptive models that improve global accuracy while maintaining performance for underrepresented samples, and validate our approach on diverse heterogeneous datasets.

A Appendix

A.1 Convergence Proof

The formal convergence proof of `FedGRAA` (Algorithm 2) is developed based on a set of standard assumptions widely adopted in the federated learning and optimization literature.

Preliminaries and Notation Let $\theta_g \in \mathbb{R}^d$ denote the global model parameters. At round t, client k has local parameters $\theta_k^t \in \mathbb{R}^d$. The aggregation weights are $w^t = [w_1^t, w_2^t, \ldots, w_K^t] \in \mathbb{R}^K$. The global validation dataset is $\mathcal{D}_\mathcal{V}$, and $\mathcal{D}_\mathcal{V}^n$ represents the n underrepresented classes within $\mathcal{D}_\mathcal{V}$.

Assumptions

Assumption 1. *[Smoothness and Convexity] The composite validation loss $L'(\theta)$ is convex and β-smooth:*

$$\|\nabla L'(\theta_1) - \nabla L'(\theta_2)\| \leq \beta \|\theta_1 - \theta_2\|, \quad \forall \theta_1, \theta_2 \in \mathbb{R}^d.$$

Assumption 2. *[Bounded Variance and Gradients] Stochastic gradients have bounded variance σ^2, and gradients are uniformly bounded by G:*

$$\mathbb{E}\left[\|\nabla L_k(\theta_k^t; b) - \nabla L_k(\theta_k^t)\|^2\right] \leq \sigma^2, \quad \|\nabla L'(\theta)\| \leq G, \quad \forall \theta \in \mathbb{R}^d.$$

Assumption 3. *[Learning Rates] Learning rates satisfy $0 < \eta, \eta' \leq \frac{1}{\beta}$.*

Assumption 4. *[Client Participation] A fixed set of K clients participate in each round, and each client possesses sufficient data to perform local updates.*

Objective Function The goal is to minimize the composite validation loss:

$$L'(\theta_g') = (1-\gamma) L_a(\theta_g'; \mathcal{D}_\mathcal{V}^h, \mathcal{D}_\mathcal{V}^m) + \gamma L_u(\theta_g'; \mathcal{D}_\mathcal{V}^n), \quad \gamma \in [0, 1].$$

Convergence Theorem

Theorem 1. *Under the above assumptions, `FedGRAA` generates sequences $\{\theta_g^t\}$ and $\{w^t\}$ such that θ_g^t converges to θ_g^*, the minimizer of $L'(\theta_g)$ as $t \to \infty$:*

$$\lim_{t \to \infty} L'(\theta_g^t) = L'(\theta_g^*) = \inf_\theta L'(\theta)$$

Proof We will prove by analyzing the updates of the parameters w^t and θ_g^t.

Weight Update via FedGRAA It updates the aggregation weights w^t using gradient descent on composite loss L' for E' epochs. The weights are updated as:

$$w^t = w^{t-1} - \eta' \nabla_{w^{t-1}} L' \left(\sum_{k=1}^{K} w_k^{t-1} \theta_k^t; \mathcal{D}_{\mathcal{V}}^h, \mathcal{D}_{\mathcal{V}}^m, \mathcal{D}_{\mathcal{V}}^n \right)$$

Lemma 1. *[Convergence of Weight Updates] Under the assumptions 1 through 3 and with $\eta' \leq \frac{1}{\beta}$, the sequence $\{w^t\}$ converges to w^* minimizing $L'(\theta_g)$.*

Proof. Since L' is convex and β-smooth 1, gradient descent with $\eta' \leq \frac{1}{\beta}$ ensures convergence to w^* 3. The update decreases L' at each step:

$$L'(w^t) \leq L'(w^{t-1}) - \left(\eta' - \frac{\beta(\eta')^2}{2} \right) \|\nabla L'(w^{t-1})\|^2$$

Choosing $\eta' \leq \frac{1}{\beta}$ ensures that RHS is non-increasing, leading to convergence.

Global Model Updates The global model is updated as:

$$\theta_g^t = \sum_{k=1}^{K} w_k^t \theta_k^t$$

Lemma 2. *[Convergence of Global Model] Given the convergence of $\{w^t\}$ to w^* and the boundedness of client updates 2, the sequence $\{\theta_g^t\}$ converges to θ_g^*, the minimizer of $L'(\theta_g)$.*

Proof. From Lemma 1, we have that $w^t \to w^*$ and local models θ_k^t are updated via gradient descent on local losses, θ_k^t approach their local minima. Therefore:

$$\theta_g^t = \sum_{k=1}^{K} w_k^t \theta_k^t \to \sum_{k=1}^{K} w_k^* \theta_k^* = \theta_g^*,$$

where θ_k^* minimizes k^{th} client's loss. The optimal weights ensure θ_g^* minimize L'.

□

Conclusion Combining the lemmas, we conclude that both w^t and θ_g^t converge to w^* and θ_g^*, respectively, ensuring that θ_g^t minimizes $L'(\theta_g)$ as $t \to \infty$.

Note

Under convexity, smoothness, bounded gradients & appropriate learning rates, `FedGRAA` ensures the convergence of aggregation weights & global weights to their optimal values, effectively minimizing the composite loss.

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